RACE
Regular Analysis Through Continuous Exercise

Mole Concept : RACE-01

Isotopes; Isobars; Isotones; Isodiaphers; Isosters; Isoelectronic (Calculation of Electron, Proton and Neutron)

1. Species which are isoelectronic to one another are

(a) $CN^{\ominus}$ (b) $OH^{\ominus}$ (c) $CH_3^{\ominus}$ (d) N_2 (e) CO

Correct answer is

(A) a, b, c (B) a, c, d (C) a, d, e (D) b, c, d

CBM080

Mole Concept (Basic Introduction to Mole; Mole Calculations)

2. The actual weight of a molecule of water is -

(A) 18 g (B) 2.99×10^{-23} g

(C) both (A) & (B) are correct (D) None of these

CBM093

3. The number of atoms in 4.25 g of NH_3 is approx :-

(A) 1×10^{23} (B) 1.5×10^{23} (C) 2×10^{23} (D) 6×10^{23}

CBM095

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4. Number of neutrons present in 1.7 g of ammonia is -

(A) N_A (B) $N_A/10 \times 4$ (C) $(N_A/10) \times 7$ (D) $N_A \times 10 \times 7$

CBM097

5. Sum of number of protons, electrons and neutrons in 12g of $^{12}_{6}C$ is :-

(A) 1.8 (B) 12.044×10^{23} (C) 1.084×10^{25} (D) 10.84×10^{24}

CBM099

6. A mixture of gases contains H_2 and O_2 gases in the ratio of 1 : 4 (w/w). What is the molar ratio of the two gases in the mixture?

(A) 4 : 1 (B) 16 : 1 (C) 2 : 1 (D) 1 : 4

CBM109

Ideal Gas Equation: Relative Density; Vapour Density (Dulong Petits Law)

7. At NTP (1 atm & 273 K), 5.6 lit of a gas weight 8 g. The vapour density of gas is :-

CBM114

Percentage Composition (Mass Percentage, Mole Percentage); Average Atomic Mass; Average Molecular Mass

8. A sample of oxygen gas (only O_2 molecules) contains O^{16} and O^{17} oxygen atoms only in 9 : 1 atomic ratio. The average molecular mass of O_2, this sample, is :

(A) 32.00 (B) 16.30 (C) 32.20 (D) 33.80

CBM289

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9. 20 gm pyrolusite ore exactly requires 29.2 gm HCl for complete reaction:

$MnO_2 + 4HCl \rightarrow MnCl_2 + Cl_2 + 2H_2O$

The mass percent of pure MnO_2 in the pyrolusite ore is (Mn = 55)

(A) 17.4 % (B) 87 % (C) 43.5 % (D) 14.6 %

CBM291

10. An elements has three isotopes X^{20}, X^{21} and X^{22}. Percentage abundance (mole %) of X^{20} is 90 and average atomic mass of X is 20.18. Select correct option(s) -

(A) Percent abudance of X^{21} = 8% (B) Percent abudance of X^{21} = 2%

(C) Percent abudance of X^{22} = 8% (D) Percent abudance of X^{22} = 2%

CBM292

Empirical and Molecular Formulae

11. An oxide of metal M has 40% by mass of oxygen. Metal M has atomic mass of 24. The empirical formula of the oxide is :-

(A) M_2O (B) M_2O_3 (C) MO (D) M_3O_4

CBM105

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	B	D	C	D	A	16	C	B	BC
Question	11									
Answer	C									

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RACE
Regular Analysis Through Continuous Exercise

Mole Concept : RACE-02

Stoichiometry With Limiting Reagent and Excess Reagents (Combustion Based Questions; Carbon Percentage; Hydrogen Percentage)

1. Ferric oxide can be obtained by oxidation of FeO:

$4FeO + O_2 \longrightarrow 2Fe_2O_3$

The O_2 gas required can be prepared by the following reaction.

$2SO_3 \longrightarrow 2SO_2 + O_2(g)$

What is the maximum amount of Fe_2O_3 can be produced by 144 g FeO and 160 g SO_3.

[Atomic mass of Fe = 56]

(A) 0.320 g (B) 80 g (C) 120 g (D) 160 g

CBM293

2. Consider the following reaction

$2Na + 2NH_3 \longrightarrow 2NaNH_2 + H_2(g)$

$2NaNH_2 + C \longrightarrow Na_2CN_2 + 2H_2(g)$

$Na_2CN_2 + C \longrightarrow 2NaCN$

Pankers Academy JEE

51gm dry ammonia gas is passed over excess heated sodium to form sodamide ($NaNH_2$) which is further reacted with carbon (excess) to finally form NaCN. Find the total volume of $H_2(g)$ evolved at 0.5 atm, 273K

(A) 201.6 L (B) 100.8 L (C) 403.2 L (D) 50.4 L

CBM294

3. $PCl_5(g)$ decomposes according to following reaction,

$PCl_5(g) \longrightarrow PCl_3(g) + Cl_2(g)$

Calculate density (in gm / *l*) of mixture at 24 atm & 273 K, when $PCl_5(g)$ undergo 50% decomposition. [R = 0.08 atm *l* mol^{-1} K^{-1}]

CBM295

4. 32.2 gm of an organic compound containing C, H & O when completely combusted produces 61.6 gm of CO_2 and 37.8 gm of H_2O. Based on this select the correct option -

(A) % of C in the organic compound is 40%

(B) % of H in the organic compound is 13.04 %

(C) % of O in the organic compound is 17.04 %

(D) Data given is impossible

CBM296

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5. Match the column-I and Column-II

	Column-I (Reactions with 2 moles A and 5 moles B)		**Column-II** (Characteristics)
(A)	A + 2B → products	(P)	'A' is limiting reagent
(B)	A + 3B → products	(Q)	'B' is limiting reagent
(C)	2A + 5B → products	(R)	During chemical reaction, mole ratio of reactants will not change
(D)	A + B → products	(S)	3 moles of excess reactant is left after complete reaction
		(T)	1 moles of excess reactant is left after complete reaction

CBM297

Principle of Atom Conservation (POAC)

6. When 6.0 gm of a hydrocarbon (M.M. = 30) is burnt completely 17.6 gm CO_2 is formed. The meolcular formula of hydrocarbon

(A) C_2H_2 (B) C_2H_4 (C) C_2H_6 (D) C_2H_6O

CBM301

7. 8 gm sulphur is burnt to form SO_2 which is further oxidised by chlorine water. The solution is then treated with $BaCl_2$ solution. The moles of $BaSO_4$ precipitated is :

[At.wt. of S = 32]

(A) 1 mole (B) 0.5 mole (C) 0.25 mole (D) 2 mole

CBM302

Percentage Yield; Percentage Purity

Bansal's Academy JEE

8. N_2O_5 decomposed according to the following sequence of reactions.

(i) $N_2O_5 \xrightarrow{100\%} NO_2 + NO_3$

(ii) $NO_3 \xrightarrow{10\%} NO + O_2$

(iii) $NO + NO_3 \xrightarrow{100\%} 2NO_2$

Calculate the number of moles of NO_2 produced due to complete decomposition of 6 mole of N_2O_5.

Fill your answer by multiplying it with 100.

CBM304

9. **Match the column :**

Column-I (Reaction)		**Column-II (At the end)**	
(A)	$\underset{4\text{ mol}}{2A} + \underset{6\text{ mol}}{2B} \xrightarrow{50\%\text{yield}} 3C$	(P)	3 moles C formed
(B)	$\underset{4\text{ mol}}{\frac{1}{2}A} + \underset{8\text{ mol}}{2B} \xrightarrow{80\%\text{yield}} C$	(Q)	3.2 moles C formed
(C)	$\underset{15\text{ mol}}{3A} + \underset{20\text{ mol}}{2B} \xrightarrow{60\%\text{yield}} C$	(R)	A is limiting reagent
(D)	$\underset{5\text{ mol}}{A} + \underset{12\text{ mol}}{3B} \xrightarrow{20\%\text{yield}} 2C$	(S)	B is limiting reagent
		(T)	1.6 moles C formed

CBM305

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Paragraph for Q.10

The following sequence of reaction occurs in commercial production of aqueous nitric acid.

$NH_3(g) + O_2(g) \longrightarrow NO(g) + H_2O(l)$ .....(1)

$NO(g) + O_2(g) \longrightarrow NO_2(g)$ .....(2)

$NO_2(g) + H_2O(l) \longrightarrow HNO_3(aq) + NO(g)$ .....(3)

In an industry 378 kg of HNO_3 was required to be produced

10. 5.00 moles of hydrogen gas, 3 moles of white phosphorus $\{P_4(s)\}$ and 12 moles of oxygen gas are taken in a sealed flask and allowed to react as follows :

$H_2(g) + P_4(s) + O_2(g) \longrightarrow H_3PO_4$

Determine the moles of ortho-phosphoric acid that can be produced, considering that the reaction occurs in 90% yield.

CBM309

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ANSWER KEY

Question	1	2	3	4	5	6
Answer	D	A	152.747	B	(A) → P,T; (B) → Q; (C) → R; (D) → P,S	C
Question	7	8	9		10	
Answer	C	720	(A) → (P,R) ; (B) → (Q,S); (C) → (P,R); (D) → (S,T)		3.00	

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RACE
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Mole Concept RACE-03

Problems based on Mixture, Sample; Impurity, and Heating Effect using Mole Concept

1. When a solid mixture of $NaHCO_3$ and Na_2CO_3 is strongly heated to 500K in an open vessel, the mass of sample decreases. The decrease is -
(A) Only due to $CO_2(g)$ formed by decomposition of Na_2CO_3
(B) Only due to $CO_2(g)$ formed by decomposition of $NaHCO_3$
(C) Only due to $CO_2(g)$ formed by decomposition of both, Na_2CO_3 and $NaHCO_3$
(D) Due to $CO_2(g)$ and water vapour formed by decomposition of $NaHCO_3$

CBM310

2. A 1800gm mixture of anhydrous $CuSO_4(s)$ and its hydrated form $[CuSO_4.5H_2O(s)]$ undergoes 20% loss in mass in heating. Mole fraction of $CuSO_4$ in mixture (Atomic mass of Cu = 64)

(A) $\frac{3}{40}$ (B) $\frac{4}{9}$ (C) $\frac{5}{9}$ (D) $\frac{1}{2}$

CBM311

3. 12.46 g of a mixture of MgO and $MgCO_3$ on strong heating lost 4.4 g in weight what is the % MgO in the mixture

CBM312

4. 1.19 gm mixture of Cu, Zn and Al contain 50% copper by weight and 3×10^{21} atoms of zinc, find gram atoms of aluminium in given mixture?
(At. mass of Al = 27 ; Cu = 63.55 , Zn=65.0) [$N_A = 6\times10^{23}$]

CBM313

5. 62.5 gm of a mixture of $CaCO_3$ and SiO_2 are treated with excess of HCl and 1.1 gm of CO_2 is produced. What is mass % $CaCO_3$ in the mixture.

CBM314

Parallel Reaction; Sequential Reaction

6. Moles of ABC_3 produced in the following set of reaction when 180 gm of A, 180gm of B and 200 gm of C are mixed in a container (given molar masses of A,B,C are 20, 30 & 10 respectively.

$2A + 3B + 5C \longrightarrow A_2BC_2 + B_2C_3$

$B_2C_3 + 3C \longrightarrow 2BC_3$

$BC_3 + A \longrightarrow ABC_3$

(A) 5 (B) 4 (C) 10/3 (D) 20/3

CBM315

Introduction; Gay Lussac's Law of Combining Volumes; Eudiometry

7. 10 mL of gaseous hydrocarbon on combustion gives 40 mL of $CO_2(g)$ and 50 mL of H_2O(vapour). The hydrocarbon is

(A) C_4H_5 (B) C_8H_{10} (C) C_4H_8 (D) C_4H_{10}

CBM111

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Molarity

8. Molarity and molality of pure CH_3COOH are respectively. (d_{CH_3COOH} = 1.5 g/ml)

(A) 16.67 , 25 (B) 25 , 16.67 (C) 50 , 33.3 (D) 25 both

CBM317

Mixing and Dilution

9. 500 mL, 0.2M Na_2SO_4 solution is mixed with 100 mL , 17.1% $\left(\frac{w}{V}\right)$ $Al_2(SO_4)_3$ solution and resulting solution is diluted to 5 times. Find the molarity of SO_4^{-2} ions in the final solution
[Mw of Al= 27, s = 32, Na = 23]

(A) $\frac{1}{12}$M (B) $\frac{5}{12}$M (C) $\frac{7}{5}$M (D) $\frac{12}{5}$M

CBM318

10. 5 ml of 1.2 M- NaOH solution is diluted till the concentration becomes 2% (w/v). Assuming volume to be additive, the volume of water added for dilution is -

(A) 12 ml (B) 7 ml (C) 19 ml (D) 5 ml

CBM320

11. 300 gm, 30% (w/w) NaOH solution is mixed with 500 gm 40% (w/w) NaOH solution. What is % (w/v) NaOH if density of final solution is 2 gm/mL :

(A) 72.5 (B) 65 (C) 62.5 (D) None of these

CBM321

12. What volume (in ml) of 0.8 M - $Ca(NO_3)_2$ solution should be mixed with 40 ml of 0.6M - $Al(NO_3)_3$ solution to get a solution having NO_3^- ion concentration equal to 1.70 M ?

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CBM322

Volume Strength of H_2O_2

13. Which of the following is incorrect for 17g/L of H_2O_2 solution?

(A) Volume strength is 5.675
(B) Molarity of solution is 0.5M
(C) 1 ml of this solution gives 2.8 ml O_2 at 273K and 2 atm
(D) The molarity of solution is 2M

CBM323

14. The volume strength of an aqueous H_2O_2 solution is '4.54V'. The concentration of solution is -

(A) 0.2 M (B) 13.6 % (w/v)
(C) 1.36 % (w/v) (D) 4.54 % (w/v)

CBM324

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	C	32.58	0.01	4.00	B	D	B	A	B
Question	11	12	13	14						
Answer	A	40	D	C						

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